

Facility-disruption 8-Ks × FIRMS: short event-vol on brief filings

Submission write-up · hypothesis · method · in-sample evidence

Massive 8-K × NASA FIRMS · page 1 of 2

+2.42%

Condor 3/10 edge vs placebo (IS)

11.5% → 5.3%

Implied move vs [realized] (median)

45 · 24 · 8

Events · IS brief · OOS brief

13%

Events where realized beat implied

1 Hypothesis & novelty

After a facility-related 8-K that FIRMS labels brief (no lasting thermal anomaly), listed options overprice the event move. The right expression is defined-risk short event-vol — an iron condor — not a long call or a stock-linked structure.

Novelty: Massive 8-K text joined to NASA FIRMS persistence, so the trade set is filings that read like plant shocks but show no heat.

Honest refinement: a fire/weather-only cut does not replicate the edge. FIRMS behaves as a negative screen on a shutdown-heavy panel, not a positive “we saw a fire from space” signal (see page 2).

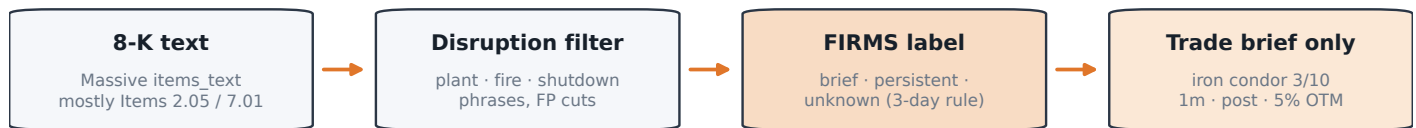
2 Method

Events 8-K items_text → disruption phrases → geocode → FIRMS → brief / persistent / unknown. Trade only brief (PERSIST_DAYS = 3).

Pricing 1m chain, entry = filing close, 5% OTM wings; synthetic spot from the ATM pair; horizons 1...21 sessions + expiry.

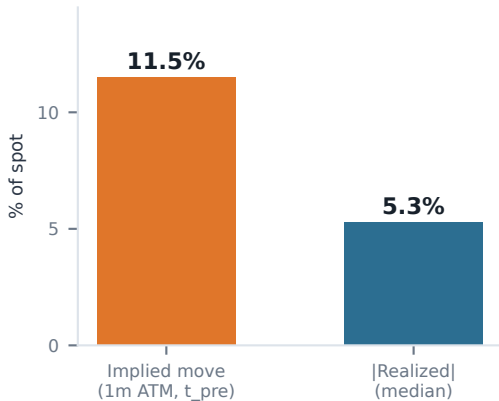
Inference Event P&L minus placebo ordinary days on the same names; bootstrap CIs; rank on h=21 + expiry; 2025 out-of-sample; 5% premium haircut; sensitivity on OTM / bucket / entry / persist.

Replication run_study(start, end) for the judges' sealed window.



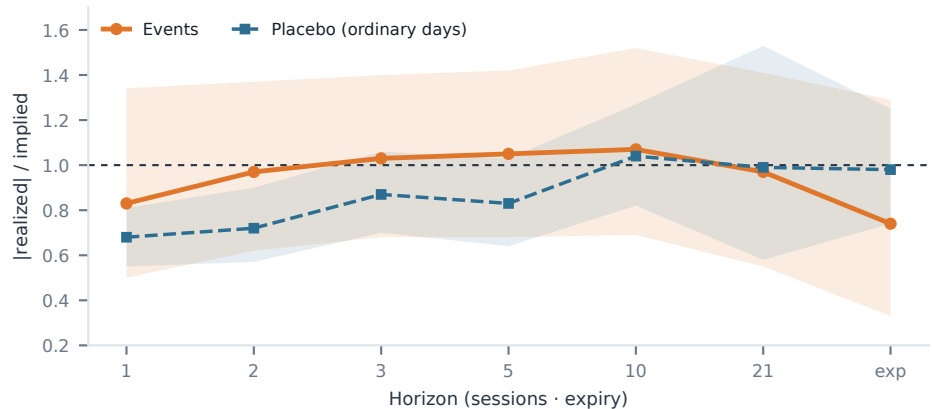
Priced vs delivered

Realized beat implied in only 13% of events



Ratio yardstick with bootstrap CIs

Below 1 ⇒ quieter than the chain priced. Event CIs still include 1 (small n)



Read

Chains price ~11% moves; paths deliver ~5%.

The core five structures are stock-linked and miss the vol edge.

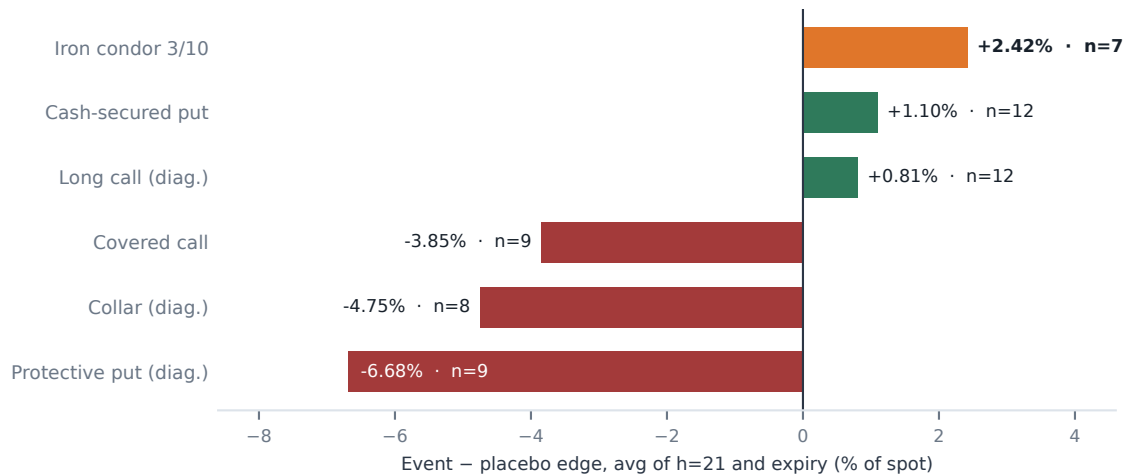
Condor 3/10 is starred vs placebo at h=10 & expiry.

Exploratory: n is small and CIs are wide.

Iron condor 3/10
Cash-secured put
Long call (diag.)
Covered call
Collar (diag.)
Protective put (diag.)

In-sample structure search · FIRMS-brief · 1m · entry = post · OTM 5%

Short premium with defined risk wins; stock-linked and long-vol structures lose vs placebo



Robustness, the fire-weather stress test, and how to trade it

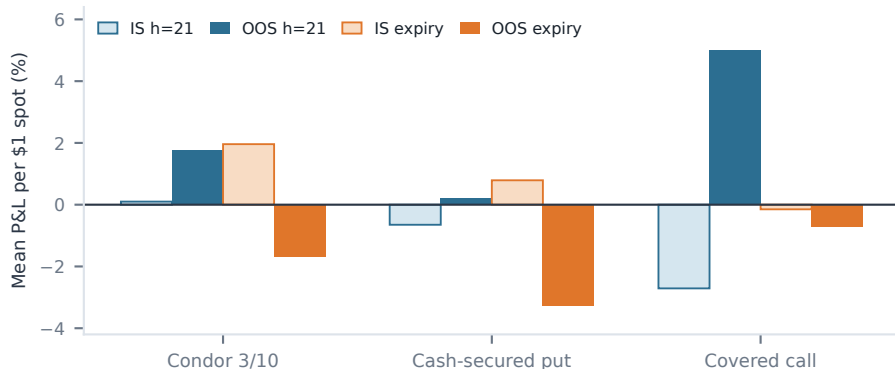
Out-of-sample · costs · acute-only cut · sensitivity · what breaks · sealed-window forecast

Massive 8-K x NASA FIRMS · page 2 of 2

3 Out-of-sample and trading costs

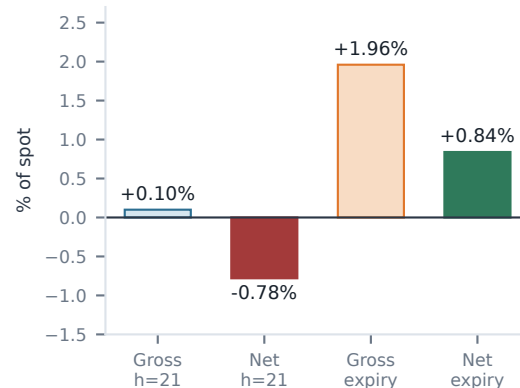
Out-of-sample transfer · 2025 · n ≈ 5 priced

Condor keeps sign mid-horizon; hold-to-expiry flips for condor and CSP



Condor 3/10 after costs

5% of premium each way · share net-positive at h=21 = 22%



4 Is the thermal signal load-bearing? Fire / weather / outage cut

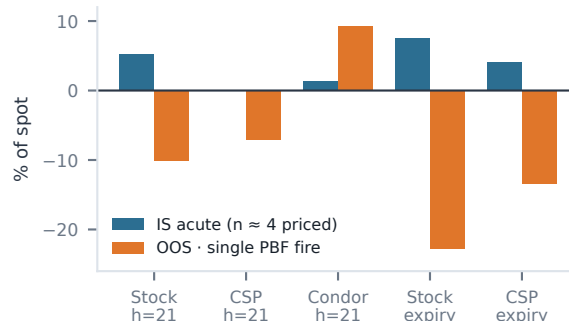
Keep only facility_fire · explosion · weather_disruption · plant_outage → 6 IS / 1 OOS events, ~4 with a usable 1m chain.

- Placebo ranking is empty at this n; the notebook falls back to “winner = long call” — not a strategy result.
- In-sample names tended to rally (stock ≈ +5% at h=21); the quiet-path story weakens.
- Sole OOS fire (PBF Martinez) moved -10% / -23%: adverse for short premium.

Conclusion The full-panel edge is mainly salient shutdown / restructuring 8-K language. Only one persistent event sits in window (VLO), so FIRMS barely reshapes the sample.

Fire / weather cut · IS vs OOS

Six IS events; the lone OOS fire moved hard



5 Parameter sensitivity

Sensitivity · mean P&L at h=21 (1m) across OTM and entry session

* = 95% CI excludes zero · entry = post unless marked pre · PERSIST_DAYS 2/3/5 leaves brief count ~unchanged (24→25)

	OTM 3%			OTM 5%			OTM 10%			pre 3%			pre 5%			pre 10%		
Long call	+2.3	+2.3	+2.3	+2.3	+2.3	+2.3	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8	+1.8
Covered call	-2.1	-2.7	-3.5	-2.1	-2.7	-3.5	-1.8	-2.3	-2.9	-1.8	-2.3	-2.9	-1.8	-2.3	-2.9	-1.8	-2.3	-2.9
Protective put	-7.0*	-7.1*	-9.1*	-7.0*	-7.1*	-9.1*	-6.5*	-6.9*	-8.4*	-6.5*	-6.9*	-8.4*	-6.5*	-6.9*	-8.4*	-6.5*	-6.9*	-8.4*
Collar	-3.7*	-4.6*	-7.9*	-3.7*	-4.6*	-7.9*	-3.5*	-4.4*	-7.2*	-3.5*	-4.4*	-7.2*	-3.5*	-4.4*	-7.2*	-3.5*	-4.4*	-7.2*
Cash-secured put	-0.8	-0.7	+1.2*	-0.8	-0.7	+1.2*	-0.7	-0.8	+1.2*	-0.7	-0.8	+1.2*	-0.7	-0.8	+1.2*	-0.7	-0.8	+1.2*
Iron condor 3/10	+0.1	+0.1	+0.1	+0.1	+0.1	+0.1	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2

6 What breaks it · how we would trade · sealed window

What would break it

- True disasters / persistent FIRMS → short vol losses; wings only bound the loss
- FIRMS false negatives contaminate brief
- Thin OTM liquidity; haircuts already erase the h=21 edge
- Treating shutdowns as satellite fires
- Automatic hold-to-expiry (OOS flips)

How we would trade it

- Only as a small, full-panel expression: iron condor 3/10, 1m, entry near filing close, exit by ~h=10-21
- Skip persistent and unknown labels
- Do not size a book on the fire/weather set — too few events, placebo broken
- Research-grade, not production

Sealed-window forecast

We predict fragility, not replication:

- Expiry P&L likely flips sign again
 - Mid-horizon condor may hold weakly
 - Any true fire in the window will hurt short premium
 - n will be single-digit; CIs will include 0
- Rubric: 30 · 30 · 20 · 10 · 10